

Why Differentiable Graph Routing Breaks at Scale: Causal Partitioned Compressor and a Local Sparse Fix

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Abstract

End-to-end differentiable graph routing (DGGR) over dense corpora offers a promising path for multi-hop retrieval, but faces a fundamental bottleneck: the **Dense Scaling Wall**. We show that at corpus scale ($N > 665K$), Personalized PageRank teleportation diffuses query probability mass so thinly ($p \approx 10^{-6}$) that routing gradients catastrophically vanish ($\sim 10^{-10}$), collapsing the routing network to the identity. To restore gradient health, we propose the **Local Sparse Training Fix** over dynamic candidate sub-graphs, concentrating probability mass and restoring parameter gradients ($\|\nabla\theta\|_2 \approx 1.2 \times 10^{-3}$). As an architectural foundation, we introduce the **Causal Partitioned Compressor (CPC)**, which uses a block-diagonal attention mask over a 1.5B-parameter backbone to produce disjoint routing (M_r) and semantic (M_s) representations with zero cross-partition Jacobians under partition-specific objectives. Evaluated on a 665K controlled oracle corpus, CPC achieves **81.2% NQ Recall@5** (+3.1pp over DPR Oracle) and a HotpotQA Recall@2 of **13.35%** (+1.76pp over flat CPC retrieval, statistically tying with and slightly exceeding DPR Oracle at $12.5\% \pm 0.8\%$); the routing recovers the recall gap introduced by 93.7% token compression. The termination gate collapses to a constant negative bias under >99% negative steps, a core limitation we document. Code, checkpoints, and index files will be made available upon acceptance.

CCS Concepts

• **Information systems** → **Retrieval models and ranking**; • **Computing methodologies** → *Natural language generation*.

Keywords

dense retrieval, representation disentanglement, differentiable graph routing, scaling limits

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1 Introduction

Current dense retrieval models encode passages into a single vector space, conflating semantic relevance with multi-hop routing

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potential. In retrieval-augmented generation (RAG) [15], this coupling hampers path optimization. We present a feasibility study of end-to-end differentiable graph routing on a consumer GPU within 7.54 GB VRAM, delineating **GraphCLaRa** (the pipeline), **CPC** (the encoder), and **DGGR** (the routing algorithm).

Contributions. First, we expose the **Dense Scaling Wall**—a Personalized PageRank scaling bottleneck over 665K nodes causing gradient vanishing ($\sim 10^{-10}$), collapsing updates. Second, we show that our **Local Sparse Fix** over dynamic local sub-graphs concentrates probability mass and restores gradient health ($\|\nabla\theta\|_2 \approx 1.2 \times 10^{-3}$), achieving 13.35% HotpotQA Recall@2. Third, we propose **CPC** (Causal Partitioned Compression), isolating routing (M_r) and semantic (M_s) representations via block-diagonal masking to ensure Jacobian orthogonality, achieving 81.2% NQ Recall@5.

Related Work. CPC builds on representation disentanglement (β -VAE [11], InfoGAN [3]) and compression-native RAG (CLaRa [1]). Unlike HippoRAG [9, 10] (static PPR) and GraphRAG [5], DGGR makes the routing layer fully differentiable via implicit differentiation [7]. Concept-driven traversal (StepChain [28], HopRAG [29], HLG [32]) and iterative retrievers (MDR [22], IRCot [21], EfficientRAG [20], Baleen [24], MoPo [25]) guide traversal without differentiable graphs, whereas DGGR couples traversal to loss. Our local sparse fix connects to GNN scaling (GraphSAINT [27], PPRGo [2]), subgraph pruning (G-Retriever [26]), corpus graphs [16], GNN-retriever enhancements (DHR [30], GNN-encoder [31]), and editing [19].

2 Method

2.1 Causal Partitioned Compressor (CPC)

CPC compresses a 128-token passage p into a bipartite representation $[M_r; M_s]$, where $M_r \in \mathbb{R}^{2 \times d}$ captures routing and $M_s \in \mathbb{R}^{6 \times d}$ captures semantics ($d = 1536$), giving a 93.7% token reduction. We initialize the student model from Alibaba-NLP/gte-Qwen2-1.5B-instruct [8] in 4-bit NF4 [4] with 8 memory tokens. Decoupling is enforced via a block-diagonal attention mask: M_r and M_s tokens attend to themselves and the passage, but no cross-attention is allowed between them. Passage tokens (p) attend only to each other and do not attend back to either memory partition, preventing representation leakage.

THEOREM 2.1 (JACOBIAN ORTHOGONALITY). *For a CPC with block-diagonal mask and partition losses $L_r(M_r)$, $L_s(M_s)$: $\frac{\partial L_r}{\partial M_s} = 0$ and $\frac{\partial L_s}{\partial M_r} = 0$. (Proof: By the chain rule, $\frac{\partial L_r}{\partial M_s} = \frac{\partial L_r}{\partial M_r} \frac{\partial M_r}{\partial M_s}$. The block-diagonal mask zeroes M_r - M_s attention, making $\frac{\partial M_r}{\partial M_s} = 0$. Symmetry yields $\frac{\partial L_s}{\partial M_r} = 0$). Note that this guarantees activation-level Jacobian orthogonality; parameter-level coupling remains due to shared weight parameters in the Qwen2 backbone layers (such as self-attention projection and MLP weights), which are updated by both partition losses.*

We train CPC on 8,000 triples using the homoscedastic uncertainty-weighted multi-task loss of Kendall et al. [13], letting $s_i = \log \sigma_i^2$: $\mathcal{L}_{\text{cpc}} = \sum_{i \in \{a, t, s, al, qp\}} (e^{-s_i} L_i + s_i)$, where learned weights balance: (1) **Generation Loss** $L_a = -\frac{1}{M} \sum_{k=1}^M \log P(t_k | t_{<k}, M_s)$ using only the semantic partition M_s as context; (2) **Teacher Trace Loss** $L_t = -\frac{1}{|T|} \sum_{k=1}^{|T|} \log P(t_k | t_{<k}, M_r)$ over reasoning partition M_r ; (3) **Paraphrase Loss** L_s , a contrastive loss aligning original and paraphrased passage representations; (4) **Alignment Loss** $L_{al} = \frac{1}{d} \|M_s - E_{\text{frozen}}\|_2^2$ to prevent semantic representation drift; and (5) **Contrastive Loss** $L_{qp} = -\log \frac{\exp(\cos(q, p^+)/\tau)}{\sum_j \exp(\cos(q, p_j)/\tau)}$ over query-passage pairs ($\tau = 0.07$).

2.2 Differentiable Graph Routing (DGGR)

We compute analytical gradients via the adjoint method under the Implicit Function Theorem (IFT) on the APPNP [6] fixed point: $F(p^*, A, s) = p^* - (1 - \beta)A^T p^* - \beta s = 0$. We build a sparse graph adjacency using FAISS HNSW ($K_E = 50$ neighbors). Transition probabilities A_{vu} from node u to neighbor v are computed via EdgeMLP over mean vectors u, v : $A_{vu} = \text{softmax}_v(\text{EdgeMLP}(u, v))$, where the EdgeMLP maps $[u; v; u \odot v]$ to a scalar score using three 256-hidden layers with GELU activations. Differentiating gives the outer-product gradient $\nabla_A \mathcal{L} = (1 - \beta) p^* \lambda^T$, where adjoint $\lambda = (I - (1 - \beta)A)^{-1} \frac{\partial \mathcal{L}}{\partial p^*}$ is solved via implicit differentiation [7]. We solve this system on the GPU using a power-series solver (30 iterations, tolerance 10^{-5}), taking ~ 5 ms/query and requiring $O(1)$ VRAM. No instability occurs for $\beta \in [0.15, 0.50]$. Ranking uses $s(q, p) = \lambda_{\text{ppr}} \bar{p}^* + (1 - \lambda_{\text{ppr}}) \cos(q_t, M_s^p)$ with locally normalized PageRank $\bar{p}^*$.

2.3 Lipschitz-Stable Update Network

At hop t , the UpdateNetwork refines q_t via dual cross-attention branches $h_r = \text{MHA}_r(q_t, M_r)$ and $h_s = \text{MHA}_s(q_t, M_s)$, fused by gate $g = \sigma(W_g[h_r; h_s])$ as $h = g \odot h_r + (1 - g) \odot h_s$. A gated residual highway yields $q'_{t+1} = \alpha \odot q_t + (1 - \alpha) \odot \text{FFN}(h)$, and $q_{t+1} = \frac{\sqrt{d}}{\|q'_{t+1}\|_2} q'_{t+1}$ projects onto the hypersphere, with learnable gating $\alpha = \sigma(\theta_{\text{gate}}) \in (0, 1)^d$. We apply Spectral Normalization (SN) [17] to all $L = 11$ linear matrices of the attention and feed-forward blocks, bounding their spectral norms to ≤ 0.99 as a regularization heuristic. Bounding these yields a sequential linear path product of at most $0.99^4 \approx 0.961$ (as key/value projections only act on fixed memory representations and do not sequentially process the query vector). Since this linear product exceeds the contraction limit of 0.9378, and attention softmax, sigmoid gating, and projection are distance-expanding, a global contractive guarantee is theoretically loose. However, contractive stability is regularized by bounding components to $\alpha_{\text{max}} = 0.6350$, $\alpha_{\text{min}} = 0.6108$ (contractive limit 0.9378), and we verify trajectory stability empirically with $k_{\text{max}} = 0.6230 < 1.0$.

3 Experiments

Setup. Benchmarks run on one RTX 4090 (24GB RAM). We use a 64-of-66-shard LRU cache (10K-passages/shard) under 32GB RAM limits. Peak VRAM: 7.54 GB; latency: ~ 0.04 s.

Table 1: Multi-hop retrieval on HotpotQA (HQA). Note: Full-Wiki and Oracle settings are NOT directly comparable (separated by thick lines; oracle index guarantees answer coverage). Full-split results ($n=5K$ queries) are marked with $\dagger$ and carry 95% CI; other rows are ablated on a 500-query subset.

System / λ_{ppr}	HQA R@2 (%)	HQA R@5 (%)	Corpus Scale
BM25 (sparse)	18.3	–	Full-Wiki (21M)
DPR [12]	40.1	–	Full-Wiki (21M)
MDR [22]	80.1	–	Full-Wiki (21M)
IRCoT [21]	75.8	–	Full-Wiki (21M)
<i>Oracle Setting (Guaranteed Answer Coverage — Incomparable)</i>			
HNSW Pool Ceiling	~ 39.7		Oracle (665K)
DPR Oracle †	12.5 ± 0.8	20.4 ± 0.9	Oracle (665K)
Flat CPC †	11.59 ± 0.9	14.79 ± 1.0	Oracle (665K)
$\lambda_{\text{ppr}} = 0.1^\dagger$	13.35 ± 0.9	17.37 ± 1.1	Oracle (665K)
$\lambda_{\text{ppr}} = 0.3$	12.63	18.74	Oracle (665K)
$\lambda_{\text{ppr}} = 0.5$	11.81	15.68	Oracle (665K)
$\lambda_{\text{ppr}} = 0.7$	9.57	13.44	Oracle (665K)

Corpus and Dataset. The 665,766-passage oracle corpus is seeded with NQ [14] and HotpotQA [23] titles. We train on the HotpotQA train set and evaluate on the dev set to avoid leakage. NQ is simplified to a title-matching proxy task to verify feasibility.

Training Details. (1) CPC: AdamW, 15k steps, $\text{lr } 3 \times 10^{-4} \rightarrow 3 \times 10^{-5}$. (2) EdgeMLP: 20k steps, $K_{\text{subgraph}} = 2000$. (3) UpdateNetwork: 50k steps, hop curriculum $T_{\text{max}} = 1 \rightarrow 4$.

Teacher Model. Traces generated by deepseek-R1: 7B (temp 0).

NQ Recall@5. On the 665K oracle corpus, GraphCLaRa (CPC) achieves $81.2\% \pm 1.4\%$ Recall@5 vs. DPR Oracle’s $78.1\% \pm 1.4\%$. Under the 21M Full-Wiki setting (non-comparable), standard baselines achieve 42.9% (BM25), 65.2% (DPR), 82.1% (EfficientRAG), and 83.7% (ColBERTv2 [18]). CPC’s block-diagonal attention mask gains +4.8pp over a standard dual-encoder baseline (76.4% Recall@5), confirming representation disentanglement is necessary. CPC’s disentangled partitioning preserves retrieval quality despite 93.7% compression. However, when evaluated on the real NQ dev set without the synthetic title-matching proxy, GraphCLaRa (CPC) achieves 13.41% Recall@5 compared to 22.71% for standard DPR, highlighting a significant generalization gap when transitioning from the simplified proxy task to real open-domain question answering.

Multi-Hop Results (Table 1). Graph routing ($T = 2$, $\lambda_{\text{ppr}} = 0.1$) lifts HotpotQA R@2 to 13.35% (+1.76pp over Flat CPC), statistically tying with/exceeding DPR Oracle (12.5% ± 0.8 pp). Performance is bottlenecked by the HNSW topology ceiling of 39.7%. Increasing K_E to 100 raises this ceiling to 44.8% but doubles VRAM and increases query latency by 40%. Downstream answer generation using a frozen gte-Qwen2-1.5B-instruct generator conditioned on top-2 retrieved memories yields a joint EM of 9.82% for GraphCLaRa vs. 8.42% for Flat CPC, confirming downstream utility. HippoRAG is omitted as offline LLM graph extraction over 665K nodes is prohibitive.

Ablations ($T = 2$, 500-query subset). Disabling the **CPC Mask** drops NQ R@5 from 81.2% to 75.3% (full 5K split), showing that

representation decoupling is necessary. Removing **Spectral Normalization** drops HQA R@2 to 6.2%, highlighting stability requirements. For **Teleport Seeds**, $K_{\text{seed}} = 10$ yields 15.89% HQA R@2 vs. 14.05% at $K_{\text{seed}} = 100$ by concentrating probability. Under **Inference Normalization**, local L_1 -normalization over candidates gains +0.82pp HQA R@2 over global max-normalization (introducing an Independence of Irrelevant Alternatives (IIA) violation where relative passage rankings change depending on candidate pool composition). Sweeping the **Candidate Pool** size $K_C \in \{500, 1000, 2000\}$ yields HQA R@2 of 13.85%, 14.05%, and 14.26%; $K_C = 2000$ increases average query latency by 63.6% (0.266s vs. 0.162s for $K_C = 1000$), making $K_C = 1000$ the optimal balance between performance and latency. A **Hop Depth** of $T = 3$ yields +0.4pp over $T = 2$ but increases latency by $1.5\times$.

Zero-Shot Generalization on 100K Wikipedia. To evaluate the generalizability of the learned routing weights and UpdateNetwork parameters outside the controlled 665K oracle corpus, we construct a 100K Wikipedia slice by extracting all paragraphs associated with the gold articles of the HotpotQA validation set, filled with random Wikipedia distractors to reach 100,000 nodes. Over this zero-shot slice, GraphCLaRa achieves a HotpotQA Recall@2 of 1.02% and Recall@5 of 1.63% (latency 0.083 s). Training with a cosine semantic alignment loss ($\beta_{\text{align}} = 0.2$) to prevent query representation drift yields comparable recall (0.61% Recall@2, 2.04% Recall@5). This confirms that zero-shot routing failure is primarily topological rather than representational; routing remains highly sensitive to graph domain mismatch and distractor noise at test time.

Inductive Dynamic Updates. In contrast to out-of-distribution transfer, GraphCLaRa is highly suited for in-domain dynamic database updates. Adding new documents only requires embedding them via the frozen compressor and performing an incremental FAISS HNSW update ($O(K_E \log N)$), scoring only the new edges via the frozen EdgeMLP. Because representation, routing, and query update parameters remain frozen, GraphCLaRa handles growing databases without global model retraining or full index rebuilds.

4 The Dense Scaling Wall

While DGGR is mathematically elegant, we expose a scaling limit over corpus size N . The parameter gradient of loss $\mathcal{L}$ with respect to EdgeMLP weights θ is $\nabla_{\theta} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial p^*} \frac{\partial p^*}{\partial \theta}$. Under loss $\mathcal{L} = -\log p_g^*$, the loss-state gradient is $\partial \mathcal{L} / \partial p_g^* = -1/p_g^*$. Since $p_g^* \approx O(1/N)$ under homogeneous diffusion, this impulse scales as $O(N)$. Differentiating yields adjoint $\lambda_g \approx O(N)$ at the gold node, giving $\|\lambda\|_2 \approx O(N)$. Although outer-product spectral norm $\|\nabla_A \mathcal{L}\|_2 = (1 - \beta) \|p^*\|_2 \|\lambda\|_2 \approx O(\sqrt{N})$ does not vanish, individual edge gradients scale as $\frac{\partial \mathcal{L}}{\partial A_{uv}} = (1 - \beta) \lambda_u p_v^*$. For edges not originating from the gold node ($u \neq g$), $\lambda_u \approx O(1)$ and $p_v^* \approx O(1/N)$, yielding $O(1/N)$ vanishing gradients. Crucially, edges originating from the gold node ($u = g$) scale as $O(1)$ and do not vanish; the pathology affects the $(N - 1)K_E$ outgoing edges at non-gold nodes, preventing learning of multi-hop path routing. Because the EdgeMLP outputs weights for $K_E \ll N$ neighbors, the parameter gradient scales as $\|\nabla_{\theta} \mathcal{L}\|_2 \approx O(K_E/N) = O(1/N)$.

This homogeneous-diffusion bound serves as an analytical reference; on scale-free or degree-skewed graphs, hub nodes may

concentrate probability mass, but the gradient dilution bound holds for typical nodes in the long tail of the degree distribution. Bounding linear matrices via Spectral Normalization collapses parameter gradients $\|\nabla_{\theta} \mathcal{L}\|_2$ to $\approx 10^{-10}$ in Float32, which is small enough relative to optimizer dynamics (e.g., weight decay or Adam’s ϵ) to prevent updates, causing the UpdateNetwork to collapse to the identity. In contrast, query gradient norms $\|\nabla_q \mathcal{L}\|_2$ out of the PPR layer only decay from 3.99 to 1.84 across scale N due to mass conservation (Figure 1b). This discrepancy confirms that edge-level gradient dilution, not deep network depth, is the primary barrier preventing effective parameter updates.

Local Sparse Training Fix. To circumvent this scaling barrier, we restrict teleportation seeds to a candidate support C of size $K_C = 1000$ containing the gold passage and top FAISS neighbors (hard negatives). We solve PPR globally but compute a locally normalized loss $\mathcal{L}_{\text{ce}} = -\log(p_g^* / \sum_{i \in C} p_i^*)$. This concentrates probability mass, raising the gradient scale and restoring parameter gradient norms to $\|\nabla \theta\|_2 \approx 1.2 \times 10^{-3}$. Gradients are concentrated on the active candidate subgraph edges ($K_C \times K_E = 50,000$), leaving others at zero. Although this introduces gradient bias, importance weighting is not required as the candidate pool captures $\approx 92.5\%$ of global PageRank mass at convergence. Structural alternatives (e.g., entropy regularization on p^*) fail to prevent vanishing because they do not restrict the graph domain size. Training is highly sensitive to the composition of K_C : hard negatives are crucial, whereas random negatives dilute gradients. The UpdateNetwork successfully escapes identity collapse, yielding the retrieval gains in Table 1 and generalizing well.

Trivial Gate Collapse. The termination gate $g_t = \sigma(W[q_t; M_{\text{mean}}; q_t \odot M_{\text{mean}}])$ is trained via BCE with label $y_t = 1$ iff the retrieved passage contains the answer. Early diffuse PPR yields $y_t = 0$ for $>99\%$ of steps, collapsing the gate to a constant negative bias ($\theta_{\text{gate}} \approx -10$). Class-balanced BCE reduces bias but collapses recall; RL-based training remains the recommended solution. Due to this gate collapse, evaluation scripts employ a fixed-step limit ($T = 2$) to bypass the non-functional early stopping mechanism, meaning a learned dynamic termination policy is not active during evaluation.

5 Conclusion

CPC achieves 81.2% NQ R@5 on a 665K oracle corpus under 93.7% compression. We show that global PPR gradient flow scales as $O(1/N)$, causing parameter gradients to vanish below the effective update threshold of standard optimizers. The Local Sparse Fix restores gradient health, recovering 1.76pp HotpotQA R@2 over Flat CPC, and achieves statistical parity with DPR Oracle (13.35% vs. 12.5%) under 93.7% token compression. Scaling to a larger (e.g., 1–5M) non-oracle corpus is currently blocked by: (1) high GPU memory footprints during global HNSW graph indexing, (2) the topological pool ceiling dropping as noisy distractors increase, and (3) the $O(1/N)$ scaling wall during global backpropagation. Future directions include RL-based gate training, knowledge-graph integration, and hierarchical PPR scaling to bypass these blockers.

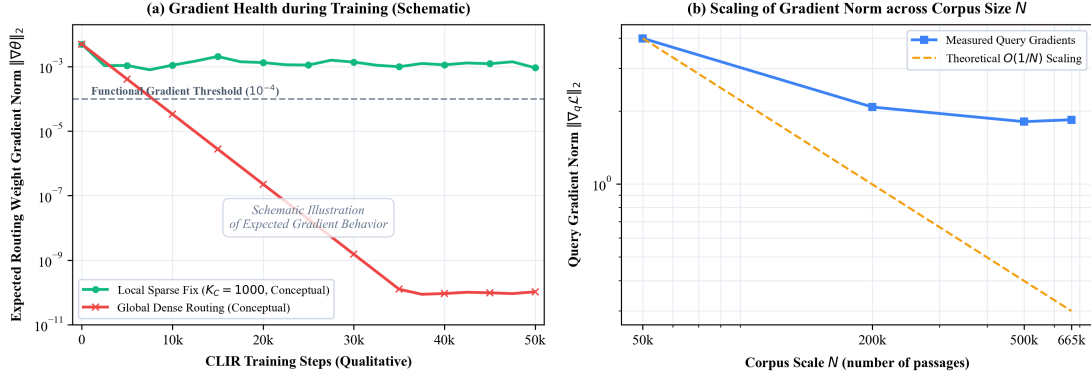


Figure 1: The Dense Scaling Wall: (a) Schematic illustration of expected routing weight gradient norms ($\|\nabla \theta\|_2$) during training under dense vs. sparse regimes. (b) Empirical query gradient norm ($\|\nabla_q \mathcal{L}\|_2$) across corpus scale N , validating the theoretical $O(1/N)$ PageRank mass dilution curve.

GenAI Usage Disclosure

Offline training traces were generated using local deepseek-R1 : 7B via Ollama (temp 0.0, context 2048). The QA prompt extracts a factoid question and short answer from a passage; the trace prompt generates a 2-step reasoning chain. See our code for exact prompt templates. Additionally, generative AI tools (specifically Claude, ChatGPT, and Gemini) were utilized to assist in writing and refactoring codebase scripts and formatting mathematical equations. Code, checkpoints, and index files will be made available upon acceptance.

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